

```
1 % calibrateSig100_invoke
2 %
3 % Modified from load_data_plot_echogram.m
4 %
5 % Dependencies:
6 %     calcEchogram70kHz
7 %     calcEchogram120kHz
8 %     calcEchogram90kHzBinned
9 %     calibrateSig100
10 %     sphere_TS
11 %     f_theorTargetStrengthBinned
12 %
13 % S. Nylund, 2019-
14 % G. Cutter, 2019-09
15
16 clear; clc; close all;
17
18 format long g;
19
20 %✓
```

----- ✓

-

```
21 %% OPTIONS
22 %✓
```

----- ✓

-

```
23 doWRITEFILE          =      true;
24 OPTIONS.doPLOTNBFigs120 = false;
25 OPTIONS.doPLOTNBFigs70  = false;
26 OPTIONS.doACCUMULATESUMMARY = true;
27
28
29 %✓
```

----- ✓

-

```
30 %% INIT
31 %✓
```

----- ✓

-

```
32 DESIRED_ECHO_FOUND = false;
33 calibration_accum = {};
```

```
34 calibration_table = table();
35
36 %✓
----- ✓
-
37 %% CALIBRATION DATA
38 % Load raw data from Sig100 converted from ad2cp to mat using ✓
MIDAS.
39 % Choose/enter inpath and infilename below.
40 % TO DO - make this UI selectable by user.
41 %✓
----- ✓
-
42 % inpath = 'D:\NortekData\ADCPE-calib-20180904\100757_Data. ✓
247.00000'
43 % infn = 'Data.247.00000.ad2cp.00000.mat'
44
45 inpath = 'D:\dev\Sig100-Calibration-data\Sig100-calibration- ✓
2019\_data_mat_from_midas'
46
47 infnlist = {...
48     'S100757A003_cal2208_0.ad2cp.00000.mat',
49     'S100757A004_cal2212_1.ad2cp.00000.mat',
50     'S100759A003_cal2208_0.ad2cp.00000.mat',
51     'S100759A004_cal2212_1.ad2cp.00000.mat',
52     'S100761A010_cal20190802.ad2cp.00000.mat',
53     'S100761A011_cal_2212_1.ad2cp.00000.mat',
54     'S100763A003_cal2208_0.ad2cp.00000.mat',
55     'S100763A004_cal2212_1.ad2cp.00000.mat',
56     'S100766A003_cal2208_0.ad2cp.00000.mat',
57     'S100766A004_cal2212_1.ad2cp.00000.mat',
58     'S100767A003_cal2208_0.ad2cp.00000.mat',
59     'S100767A004_cal2212_1.ad2cp.00000.mat',
60     'S101132A006_cal2208_1.ad2cp.00000.mat',
61     'S101132A007_cal2212_1.ad2cp.00000.mat',
62     'S101134A006_cal2208_1.ad2cp.00000.mat',
63     'S101134A007_cal2212_1.ad2cp.00000.mat'
64 }
65
66 % % NOTE, these files represent:
67 % % Calibration of 8 AMLR Nortek Sig100s using deployed firmware ✓
```

2208_0 (for 6 off Cape Shirreff), and 2208_1 (for IMR-deployed two), and ✓
new firmware 2212_1.

```
68 % % Copies of data are in
69 % % D:\NortekData\SIG100-CALIBRATION\Sig100-calibration-2019
70 % % D:\dev\Sig100-Calibration-data\Sig100-calibration-2019
71 % % \\swc-storage3-s\AERD\Moorings\Sig100\Calibration\Sig100-✓
calibration-2019
```

```
72
73 % change working directory
74 dir(inpath)
75
76
77 %✓
```

```
----- ✓
-
78 %% Load data, define water and sphere, and process (calibrate)
79 %✓
```

```
----- ✓
-
80 for(indf = 1 : length(infnlist))
81
82     close all;
83
84     infn = infnlist{indf};
85
86     %
87     f_disp_dbstack_info()
88
89     %✓
```

```
-----
90     %% Load infile
91     %✓
```

```
-----
92     load(fullfile(inpath,infn))
93     % check
94     if(exist('Data','var'))
95         fprintf('Data loaded\n\n');
96     else
97         fprintf('Data not found. Returning\n\n');
98         return;
99     end
```

```
100
101    % % File should contain Config and Data structs
102    % % TO DO - check for Config & Data.
103
104
105    %✓
-----
106    % Water & target sphere properties, and theoretical TS✓
parameters
107    %✓
-----
108
109    %% Define: Water Properties
110    WaterProp.soundVel          = 1500.0; % (m/s)
111    WaterProp.temp              = 14.0;   % (C)
112    WaterProp.sal               = 34.0;   % (psu)
113    WaterProp.density           = 1033.0; % kg/m^3
114
115    %% Define: Target sphere properties
116    sphereProp.sphereDiameter   = 25.4; % mm
117    sphereProp.sphereType       = 'WC'; % 'WC' or 'CU'✓
currently supported
118    sphereProp.approxDist       = 4.6;   % m
119
120    %% Define: Theoretical TS model parameters for 90 kHz FM mode
121    TS_theor_paramsFM.Ftminspec = 68.4; % kHz minimum✓
frequency to model
122    TS_theor_paramsFM.Ftmaxspec = 113.4; % kHz maximum✓
frequency to model
123    TS_theor_paramsFM.nF        = 500;   % number of✓
frequencies to model
124    TS_theor_paramsFM.sphereDiameter = sphereProp.sphereDiameter; %✓
25.4; % sphere diameter (mm)
125    TS_theor_paramsFM.sphereType   = sphereProp.sphereType; %✓
sphere type, 'WC' or 'CU'
126    TS_theor_paramsFM.HZPASS      = -0.1; % avoids TS samples✓
affected by filter edges
127    TS_theor_paramsFM.soundVel    = WaterProp.soundVel; % 1500 m/s✓
% water sound speed
128    TS_theor_paramsFM.rhow         = WaterProp.density; % water✓
density 1033.0
```

```
129     TS_theor_paramsFM.doPLOTTSBINNED = false;
130
131     %% Define: Theoretical TS model parameters
132     TS_theor_params70.Ftminspec      = 69.5;  % kHz minimum✓
frequency to model
133     TS_theor_params70.Ftmaxspec      = 70.5;  % kHz maximum✓
frequency to model
134     TS_theor_params70.nF              = 10;    % number of✓
frequencies to model
135     TS_theor_params70.sphereDiameter = sphereProp.sphereDiameter; %✓
25.4; % sphere diameter (mm)
136     TS_theor_params70.sphereType     = sphereProp.sphereType; %✓
sphere type, '%WC' or 'CU'
137     TS_theor_params70.HZPASS         = -0.1;   % avoids TS samples✓
affected by filter edges
138     TS_theor_params70.soundVel       = WaterProp.soundVel; %1500; %✓
water sound speed
139     TS_theor_params70.rhow           = WaterProp.density; % water✓
density 1033.0
140     TS_theor_params70.doPLOTTSBINNED = false;
141
142     %% Define: Theoretical TS model parameters
143     TS_theor_params120.Ftminspec     = 119.5;  % kHz minimum✓
frequency to model
144     TS_theor_params120.Ftmaxspec     = 120.5;  % kHz maximum✓
frequency to model
145     TS_theor_params120.nF            = 10;    % number of✓
frequencies to model
146     TS_theor_params120.sphereDiameter = sphereProp.sphereDiameter; %✓
25.4; % sphere diameter (mm)
147     TS_theor_params120.sphereType    = sphereProp.sphereType; %✓
sphere type, '%WC' or 'CU'
148     TS_theor_params120.HZPASS        = -0.1;   % avoids TS samples✓
affected by filter edges
149     TS_theor_params120.soundVel      = WaterProp.soundVel; %1500; %✓
water sound speed
150     TS_theor_params120.rhow          = WaterProp.density; % water✓
density 1033.0
151     TS_theor_params120.doPLOTTSBINNED = false;
152
153     %% Struct with combined TS theor params
```

```
154     TS_theor_params.TS_theor_paramsFM = TS_theor_paramsFM;
155     TS_theor_params.TS_theor_params70 = TS_theor_params70;
156     TS_theor_params.TS_theor_params120 = TS_theor_params120;
157
158
159
160     %✓
-----
161     %% CALIBRATION
162     % % calibrateSig100 function call
163     %✓
-----
164     f_disp_dbstack_info()
165     fprintf('Calling calibrateSig100().\n');
166
167     close all
168     [spherePos,TS_theor_binned,calibration] = calibrateSig100✓
(Config,Data,sphereProp,WaterProp,TS_theor_params);
169
170     % % % % TS_theor_binned returned contains
171     % % TS_theor_binned.TS_theor_binned90fm = TS_theor_binned90fm;
172     % % TS_theor_binned.TS_theor_binned70    = TS_theor_binned70;
173     % % TS_theor_binned.TS_theor_binned120   = TS_theor_binned120;
174
175
176     %✓
-----
177     %% Calibration Results
178     %   Call function to display results and optionally write to✓
file
179     %✓
-----
180     disp('calibration struct: '); disp(calibration.fm90);
181
182     f_disp_dbstack_info()
183     fprintf('Calling f_report_calibration() to write results.\n');
184
185     % Call function to write calibration results to console and✓
textfile (optional)
186     f_report_calibration(calibration,Config,doWRITEFILE);
187
```

```
188
189 % SAVE RESULTS TO MAT FILE
190 outpath = fullfile( pwd, 'Results');
191 if(~isdir(outpath))
192     mkdir(outpath);
193 end
194 %ofn=fullfile(outpath,'calibration_summary_sig100.mat');
195 %% Save calib summary to mat file, default filename✓
('CAL_ser#_fw#.mat')
196 [calibration_summary_indiv] = f_save_calibration(calibration,✓
Config,[]); %
197
198 %if(OPTIONS.doACCUMULATESUMMARY)
199 calibration_accum = [ calibration_accum ;✓
calibration_summary_indiv.calarray];
200 calibration_table = [ calibration_table ;✓
calibration_summary_indiv.caltable];
201 %end
202
203 end
204
205
206
207 %% Save combined calibration data table to mat file
208 save( fullfile( outpath, 'Sig100_calibration_combined.mat'),✓
'calibration_accum');
209 save( fullfile( outpath, 'Sig100_calibration_table.mat'),✓
'calibration_table');
210
211 %% Write combined calibration data table to text file
212 ofncalcombined = 'Sig100_calibration_combined.txt';
213 doDEFAULTOFN = true;
214 [fidout,outfn] = f_openfile( ofncalcombined , doDEFAULTOFN);
215 hdr='';
216 fmt = calibration_summary_indiv.calheaderfmt;
217 for(indh = 1 : length(calibration_summary_indiv.calheader))
218     hdr = sprintf('%s%s,', hdr, calibration_summary_indiv.calheader✓
{indh});
219 end
220 hdr(length(hdr))=[]; %fprintf('HEADER: %s\n',hdr);
221 fprintf(fidout, '%s\n',hdr);
```

```
222 rowstr = '';
223 for(indr = 1 : size(calibration_accum,1))
224     %fprintf(fidout,'%s,',calibration_accum{indr,indc});
225     rowstr = sprintf('%s%d,', rowstr, calibration_accum{indr,1});
226     rowstr = sprintf('%s%d,', rowstr, calibration_accum{indr,2});
227     rowstr = sprintf('%s%s,', rowstr, calibration_accum{indr,3});
228     rowstr = sprintf('%s%.2f,',rowstr, calibration_accum{indr,4});
229     rowstr = sprintf('%s%.4f,',rowstr, calibration_accum{indr,5});
230     rowstr = sprintf('%s%s', rowstr, calibration_accum{indr,6});
231
232     fprintf(fidout,'%s\n',rowstr);
233     rowstr='';
234 end
235 fclose(fidout);
236
237 fclose all;
238
239 % END
240 fprintf('\nFunction calibrateSig100_invoke completed.\n');
241
242
243
244
```